

UNIVERSITY OF WATERLOO
CS247 – Software Engineering Principles
Project Specification
RuneVault
The Arcanist's Descent

Instructor:	Victoria Sakhnini
Course:	CS247 – Software Engineering Principles

Important: DO NOT SUBMIT WITHOUT COMPILING AND TESTING FIRST. If your final submission does not compile or otherwise does not run, you will have nothing to show during your demo. Resist the temptation to make last-minute changes; they are almost certainly not worth the risk.

Above all, **MAKE SURE YOUR SUBMITTED PROGRAM RUNS.** Markers do not have time to read source code and award partial marks for non-working programs. Even a partially working program that runs is far better than one that does not.

The Game of RuneVault

In this project, you will implement **RuneVault** (RV), a single-player dungeon-crawler in the tradition of classic rogue-like games. You play as an Arcanist, a scholar of forbidden magic who has descended into the Cursed Vault beneath the Academy, seeking the five Shards of the Shattered Seal. Each Shard is locked on a different vault level and guarded by increasingly powerful Spectral Wardens.

A game of RuneVault consists of a board **79 columns wide and 30 rows tall** (5 rows are reserved for displaying status information). The Arcanist moves through five vault levels, dispelling wards, collecting rune fragments, and defeating spectral enemies until claiming the Shard on level 5 or being overwhelmed by the Vault's defences. RuneVault redraws the entire board after every command rather than updating in real time.

Some Definitions

- **Arcanist (AC):** The player character, controlled by the player. Always shown as @ on the board.
- **Spectre:** A non-playable enemy character inhabiting the vault. All Spectres are hostile unless noted otherwise.
- **Rune Fragment (RF):** The primary currency of RuneVault. Collected by walking over it or defeating certain enemies.
- **Ward:** A magical lock on the descent to the next level. Only lifted when the Arcanist carries the Cipher Gem for that level.

- **Vault Level:** A predefined configuration of 5 chambers connected by arcane corridors. The configuration is the same on every level.
- **Focus Points (FP):** The representation of the Arcanist's magical reserves, equivalent to HP. When FP reaches 0, the run ends.
- **Power (Pwr):** The representation of a character's offensive strength.
- **Ward Rating (WR):** The representation of a character's magical resistance, equivalent to Def.
- **Cell:** Either a vault wall, floor tile, archway, or corridor segment.
- **Spawned:** An enemy, fragment, or scroll has been placed on the board at the start of the level.
- **1 Block Radius:** The 8 cells immediately adjacent to a character or item.
- **Shard:** A Shard of the Shattered Seal, represented in play as a Shard Coffin (see Rune Fragments). Exactly one Shard is placed on every vault level, guarded by a Vault Anchor.

System Components

Arcanist Classes

By default, the Arcanist plays as a **Sage** (120 FP, 18 Pwr, 18 WR). At the start of each game, the player selects one of the four classes below. The Arcanist is always shown as @ on the board.

Class	FP	Pwr	WR	Class Ability
Sage (s)	120	18	18	Rune Fragments collected are worth +50% at end-game scoring
Hexblade (h)	100	24	14	All scrolls produce double their normal effect (positive and negative)
Warden (w)	150	16	22	Damage taken from Spectres is reduced by 1 (minimum 1)
Voidwalker (v)	110	22	12	Negative scrolls have a chance (50%) to produce their positive counterpart instead

Design Question 1:

How could you design your system so that each Arcanist class can be easily generated, and its class ability cleanly applied without scattering conditionals throughout the codebase? How difficult does such a design make it to add a fifth class later?

Spectres

Spectres are the Arcanist's adversaries inside the vault. Unless noted otherwise, Spectres move one square randomly within the chamber in which they were spawned. The exception is **Vault Anchors**, who are stationary and always guard a Shard Coffin.

Specter	Symbol	FP	Pwr	WR	Special
Wraith	W	60	20	10	—
Banshee	B	40	30	5	—
Revenant	R	100	18	20	—
Shade	S	50	8	8	—
Lich	L	30	40	5	Becomes permanently hostile to all future Arcanists if attacked
Vault Anchor	A	120	22	18	Always guards a Shard Coffin; stationary
Specter Lord	X	80	28	15	—

Lich: A Lich can be attacked and defeated by the Arcanist. Attacking or slaying a Lich causes every Lich on subsequent levels to become hostile, attacking the Arcanist if it passes within a one-block radius.

Vault Anchors: A Vault Anchor always spawns within a one-block radius of its Shard Coffin. Whenever a Shard Coffin is placed on a level, a Vault Anchor is also placed to guard it. The Vault Anchor is only hostile when the Arcanist is adjacent to the Shard Coffin.

Spectre drops: Upon defeat, any Spectre that is not a Vault Anchor or Lich drops 1 Rune Fragment, immediately added to the Arcanist's total.

Spectre movement: Spectres (except Vault Anchors) move randomly one floor tile at a time, provided the tile is unoccupied. A Spectre can never leave the chamber in which it was spawned. Spectres are moved row by row, left to right. Any particular Spectre is moved at most once per Arcanist action.

Spectre attacks: If the Arcanist is within a 1-block radius of a hostile Spectre, that Spectre attacks instead of moving. There is a 50% chance the attack misses.

Design Question 2:

How does your system handle generating the different Spectre types? Is the approach different from how you generate the Arcanist class? Why or why not?

Design Question 3:

How could you implement special abilities for Spectres, for example, Focus Point drain for Wraiths, temporary Warding reduction for Banshees, or rune-fragment theft for Shades? What design pattern best models abilities that modify combat outcomes without permanently altering the character objects involved?

Items

Arcane Scrolls

Scrolls are the consumable items of RuneVault. All scrolls are shown as ? on the board. The Arcanist cannot carry scrolls; a scroll can only be used when the Arcanist is within a 1-block radius of it. The effect of a scroll is not known until it is first used in a session. After first use, that scroll's type is remembered for the remainder of the run; other scroll instances remain unknown.

Type	Abbrev.	Effect	Duration
Attune Focus	AF	Restore up to 10 FP (cannot exceed class maximum)	Permanent
Surge Power	SP	Increase Pwr by 5	Current level only
Fortify Ward	FW	Increase WR by 5	Current level only
Drain Focus	DF	Lose up to 10 FP (cannot fall below 0)	Permanent
Sap Power	SaP	Decrease Pwr by 5 (cannot fall below 0)	Current level only
Erode Ward	EW	Decrease WR by 5 (cannot fall below 0)	Current level only

Design Question 4:

What design pattern could you use to model the temporary effects of Surge Power, Fortify Ward, Sap Power, and Erode Ward, so that you do not need to explicitly track which scrolls the Arcanist has used on a given level? Consider how level transitions should trigger the reversal of these effects.

Rune Fragments

Rune Fragments are the scoring currency of RuneVault, shown as * on the board. They come in several grades:

Fragment Type	Value	Notes
Common shard	1	Standard drop
Resonant shard	2	Standard drop
Lich cache	4	Dropped on the defeat of a Lich
Shard Coffers	6	Guarded by a Vault Anchor; can only be collected after the Anchor is defeated

Major Items

Two unique items appear across the run, one recurring and one unique.

The Cipher Gem (C): Appears once per level. At the start of each level, one Spectre is chosen uniformly at random to carry the Cipher Gem, excluding Vault Anchors. When that Spectre is defeated, the Gem drops on its tile and is picked up by walking over it. Picking up the Cipher Gem lifts the Ward on the descent to the next level and reveals the stairway (shown as \) on the display.

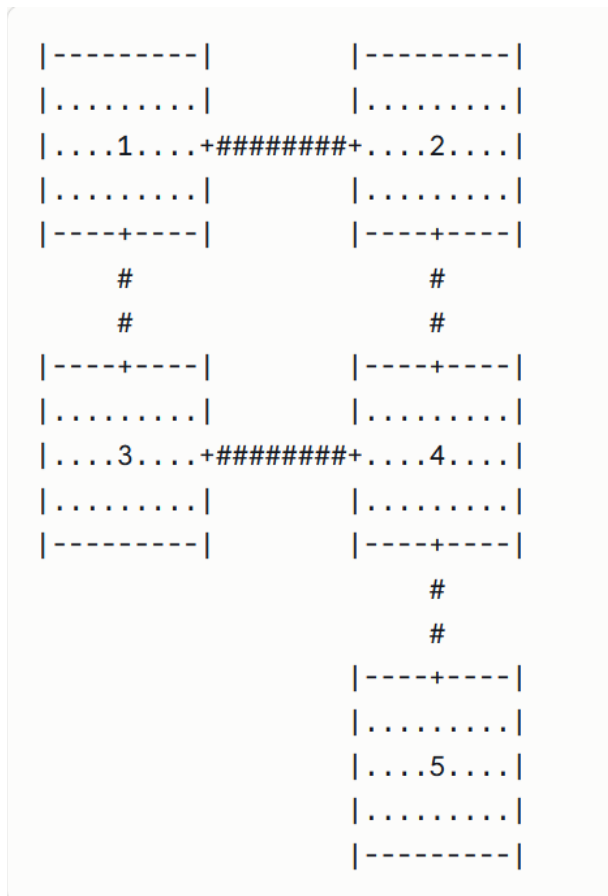
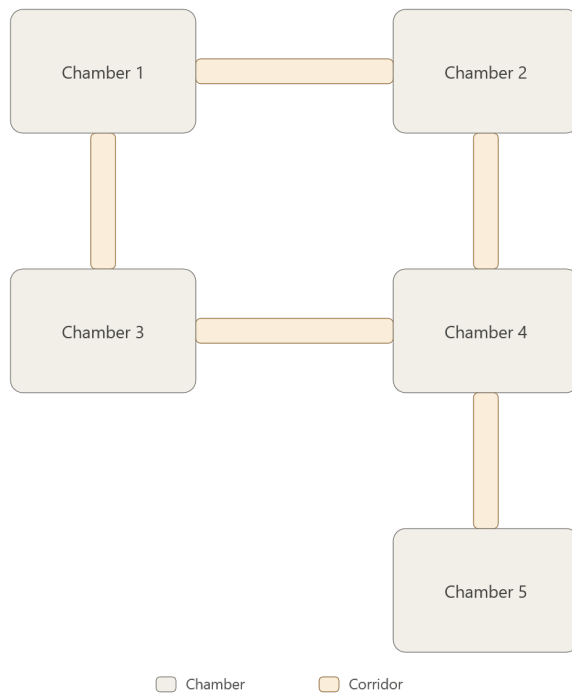
The Aegis Cloak (G): Appears only once per run, on a randomly selected level (2–4). It follows the same placement rules as a Shard Coffin, always guarded by a Vault Anchor and inaccessible until that Anchor is defeated. While the Arcanist carries the Aegis Cloak, all damage taken is halved (rounded up).

Design Question 5:

How could you design item generation so that Rune Fragments, Scrolls, and Major Items reuse as much code as possible? How would you structure your hierarchy to avoid duplicating the 'guarded placement' logic shared by the Shard Coffin and the Aegis Cloak?

Vault Levels

Each vault level consists of 5 chambers connected in the same fixed configuration (see Figure 1). All five levels share this layout. The chambers are connected as follows: chamber 1 connects to chambers 2 and 3; chamber 2 connects to chamber 4; chamber 3 connects to chamber 4; and chamber 4 connects to chamber 5. Every chamber is therefore reachable from every other chamber. A corridor always meets a chamber at an archway (+). The Ward on the descent is lifted only when the Arcanist carries the Cipher Gem for that level.



Spawning Rules

Arcanist: Spawns randomly on a floor tile in any chamber. The Arcanist must not spawn in the same chamber as the descent point (stairway). Chamber selection and tile selection within the chamber are uniformly random.

Descent (stairway): Shown only after the Arcanist picks up the Cipher Gem. Without the Gem, the stairway tile looks like a regular floor tile, but walking over it has no effect until the Ward is lifted.

Scrolls: 10 scrolls spawned per level, each of the 6 types with equal probability (1/6). Any chamber has an equal probability of hosting a scroll; any floor tile within the selected chamber is equally likely.

Rune Fragments: Exactly one Shard Coffin (the level's Shard) is spawned per level, guarded by a Vault Anchor. The remaining 9 Rune Fragment piles are common shards (probability 5/7) or resonant shards (probability 2/7). Chambers and floor tiles within chambers are equally likely.

Spectre Spawn Distribution:

20 Spectres spawned per level. Every chamber is equally likely to spawn any particular Spectre. Spawn probabilities (excluding Vault Anchors):

Specter	Probability
Wraith	2/9
Banshee	3/18
Shade	5/18
Revenant	1/9
Specter Lord	1/9
Lich	1/9

Generation order: Arcanist location, stairway location, scrolls, rune fragments, spectres. This ordering allows for consistent evaluation of random generation.

Multiple objects cannot occupy the same cell. No two objects may ever share a space, except rune fragments, which are collected when the Arcanist walks over them. Spectres and items may only spawn on floor tiles, never in archways, corridors, or on the stairway.

Combat

All Spectres except Liches and Vault Anchors are hostile by default. If the Arcanist enters within a 1-block radius of a hostile Spectre, the Spectre attacks instead of moving. Vault Anchors are only hostile when the Arcanist is adjacent to their Shard Coffers. Spectres never pursue the Arcanist.

Initiative: The Arcanist always has initiative and attacks first.

Spectre attacks: Auto-attack when the Arcanist is within a 1 block radius. There is a 50% chance the attack misses.

Arcanist attacks: The Arcanist may attack any Spectre within a 1-block radius. The Arcanist never misses.

Damage Formula

$$\text{Damage(Defender)} = \text{ceiling}((100 / (100 + \text{WR}(\text{Defender}))) \times \text{Pwr}(\text{Attacker}))$$

where Attacker is the attacking character, and Defender is the character being attacked.

If a Spectre's attack hits, the Arcanist's FP is reduced by Damage(AC) as calculated above (halved if the Aegis Cloak is carried).

Aegis Cloak effect: If the Arcanist carries the Aegis Cloak, the damage they take is halved (rounded up):

$$\text{Damage(AC)} = \text{ceiling}(\text{Damage(Defender)} / 2)$$

If both the Warden's -1 reduction and the Aegis Cloak apply, resolve them in this order: subtract the Warden's 1, halve the result rounding up, then apply the minimum of 1.

$$\text{Damage(AC)} = \max(1, \text{ceiling}((\text{Damage} - 1) / 2))$$

Example: base damage 4 $\rightarrow \max(1, \text{ceiling}(3 / 2)) = 2$.

Display

The board redraws completely after every command. The bottom 5 rows display status information. The Arcanist can see all chambers simultaneously.

Symbol	Meaning
@	Arcanist (player character)
, -	Vault walls
+	Archway (doorway)
#	Corridor
.	Floor tile (walkable)
\	Descent to next level (only visible after picking up the Cipher Gem)
S	Arcane Scroll (type unknown until used)
*	Rune Fragment (all types)
?	Cipher Gem (drops when its carrier is defeated)
G	Aegis Cloak
W	Wraith
B	Banshee
R	Revenant
S	Shade
L	Lich
A	Vault Anchor
X	Specter Lord

Recommendation: Use ANSI colour output codes to make the Arcanist and descent point stand out. Use different colours for Spectres, Rune Fragments, and Scrolls to improve readability.

Command Interpreter

Initially, the game will prompt the player to select an Arcanist class or quit. Play then continues until the Arcanist reaches the end of level 5, their FP reaches 0, the player restarts, or the player quits. If the run ends, the player is offered the option to play again.

The following commands are supported:

- **no, so, ea, we, ne, nw, se, sw** , Moves the Arcanist one cell in the specified cardinal or diagonal direction.
- **u direction**, Uses the scroll in the specified direction (e.g., u no, u ea). The scroll must be within a 1-block radius.
- **a direction: Attacks the Spectre** in the specified direction. The Spectre must be immediately adjacent. If there is no Spectre adjacent in the specified direction, the attack command has no effect, and the turn is not consumed
- **s, h, w, v** , Selects the Arcanist's class at game start (Sage, Hexblade, Warden, Voidwalker).
- **r** , Restarts the run. All stats, scrolls, and rune fragments are reset. A new class must be selected.
- **q** , Quits the game.

The board redraws after every command.

Ending the Run and Scoring

A run ends in one of the following ways:

- The Arcanist reaches and descends the stairway on level 5 (win).
- The Arcanist's FP reaches 0 (loss).
- The player restarts the run.

A score is generated only on a win. The score equals the total Rune Fragments collected, except that Sages receive a 50% bonus to their final score.

Command-Line Options

Your program should support the following optional command-line argument:

- **Level layout file:** A single optional argument specifying a file that defines the layout of all five levels. The file provides the exact level layout as presented to the player (minus the 5 status rows at the bottom). Scrolls and rune fragments are represented by numbers in the file, which are converted to the appropriate characters on display.

Number	Item
0	AF (Attune Focus scroll)
1	SP (Surge Power scroll)
2	FW (Fortify Ward scroll)
3	DF (Drain Focus scroll)
4	SaP (Sap Power scroll)
5	EW (Erode Ward scroll)
6	Common shard
7	Resonant shard
8	Lich cache
9	Shard Coffers

Optional: You may support a second optional argument specifying a seed for the random number generator, enabling reproducible runs. This is not required.

Grading & Submission

Submission Requirements

Your project will be evaluated through the following deliverables:

- Code: Submitted on Marmoset. Make sure your program compiles and runs before submitting.
- Demo: A live demonstration of your working program to the course staff.
- Report: See the separate Phase 1 and Phase 2 Report Requirement documents.

If Things Go Wrong

If time is running short, here is the suggested order of priorities:

1. Reading and displaying a vault level with Spectres, scrolls, rune fragments, and descent point at specified locations.
2. A general-purpose Arcanist, Spectre enemies, and items (no special types yet).
3. Movement and interaction (combat, scroll use, fragment collection) for the generalized versions.
4. Different Arcanist classes and Spectre types.
5. Different scroll and rune fragment types.
6. Random level generation.

Grading note: Accomplishing the first three points should earn at least 50%, assuming good object-oriented principles. A program that runs but is incomplete always receives a higher mark than one that attempts everything but does not compile.

If Things Go Well (Bonus Features)

If you complete the full specification, you may implement bonus features for extra credit. You must support play both with and without your bonus features; recompilation with different flags is not permitted.

- Curses library: Use a curses/ncurses library for real-time WASD controls.
- Inventory system: Allow the Arcanist to carry multiple scrolls; add new relic-type items.
- Spectre intelligence: Give Spectres the ability to pursue the Arcanist across corridors.
- Lich trading: Allow Liches to exchange rune fragments for scroll hints before hostilities are declared.
- Prestige system: Add a class upgrade mechanic (e.g., Sage → Archmage) that unlocks after reaching level 3.
- Additional classes, Spectres, or items that pose interesting OOP design challenges.

Note: To earn significant credit, enhancements must be algorithmically complex or solve an interesting OOP design problem. Trivial additions will not earn a significant fraction of available marks.

A Note on Random Generation

This project requires pseudo-random number generation. See the provided `shuffle.cc` for an example of seeding and using a random number generator. The generator should be seeded once at startup. Runs started with the same seed will produce identical sequences of random events.